

A PROJECT DOCUMENTATION

ON

**SAP PROCURE-TO-PAY (P2P) INTEGRATION
USING REST APIS**

SUBMITTED BY

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PROGRAM : SAP INTEGRATION DEVELOPER (IE)



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PROBLEM STATEMENT:

Organizations rely heavily on procurement processes to purchase goods and services from vendors. Traditional procurement methods involve manual approvals, documentation, and communication, which often result in delays, errors, and inefficiencies.

SAP systems provide automation for procurement through Procure-to-Pay (P2P) processes, but understanding SAP integration requires practical implementation.

This project aims to simulate the SAP Procure-to-Pay integration process using REST APIs. The system automates vendor selection, purchase requisition, purchase order creation, goods receipt, invoice verification, and payment processing.

PROJECT OBJECTIVE:

The main objectives of this project are:

- To understand SAP Procure-to-Pay business process
- To simulate SAP integration workflow
- To implement REST APIs for business processes
- To demonstrate end-to-end procurement automation
- To gain hands-on experience in SAP Integration development

SOLUTION / FEATURES:

To address the challenges in manual procurement systems, this project implements a simulated SAP Procure-to-Pay (P2P) Integration using REST APIs. The solution automates the entire procurement lifecycle from vendor selection to payment processing. The system is developed using Node.js and Express.js, where each stage of the Procure-to-Pay process is represented as an API endpoint. These APIs interact with each other to simulate real-world SAP integration between procurement and finance modules.

The workflow begins with retrieving vendor details, followed by creating a purchase requisition. After approval, a purchase order is generated and sent to the vendor. Once the goods are received, invoice verification is performed. Finally, the payment is processed, completing the procurement cycle.

This approach provides a simplified simulation of SAP integration while maintaining the structure and logic of real-world business processes.

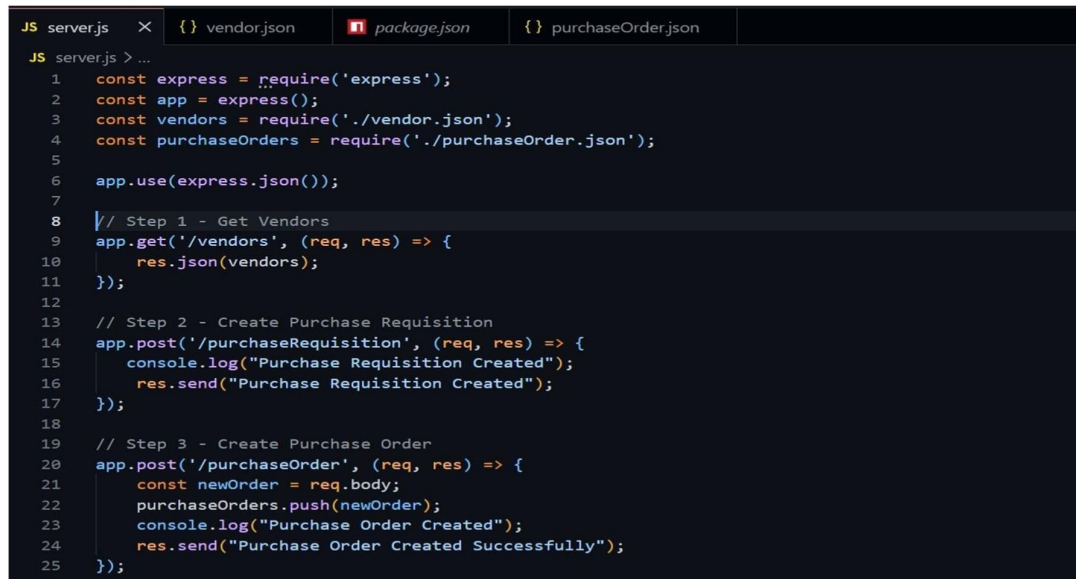
■ KEY FEATURES:

- Vendor Management
- Purchase Requisition Creation
- Purchase Order Processing
- Goods Receipt Confirmation
- Invoice Verification
- Payment Processing

- REST API Based Integration
- End-to-End Procure-to-Pay Workflow

SCREENSHOTS:

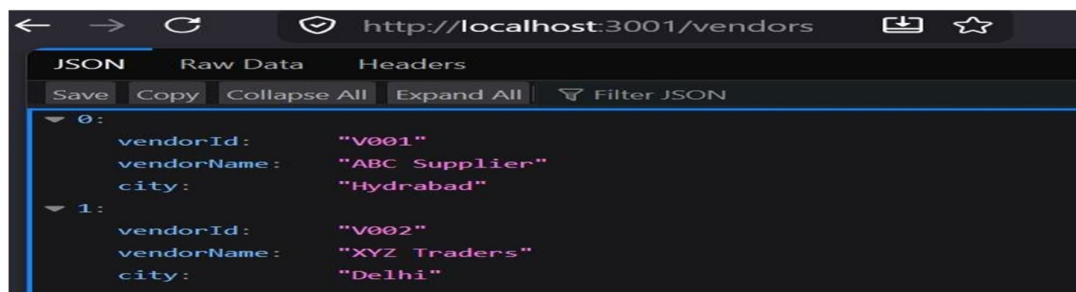
■ End-to-end-P2P Integration:



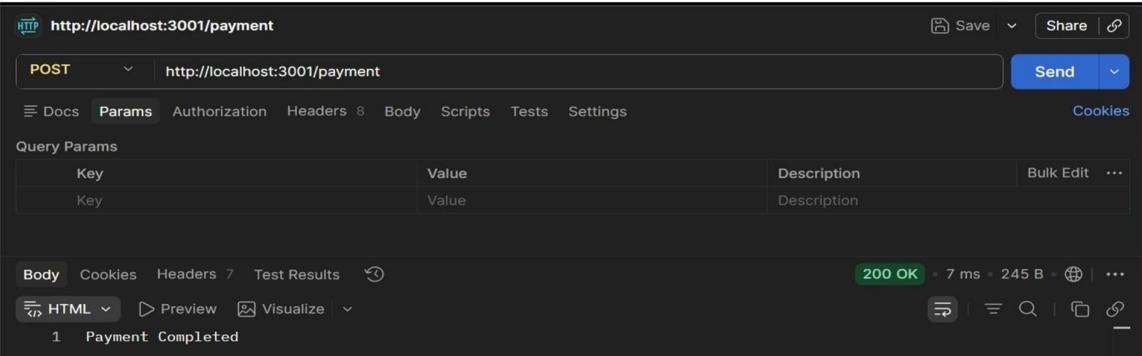
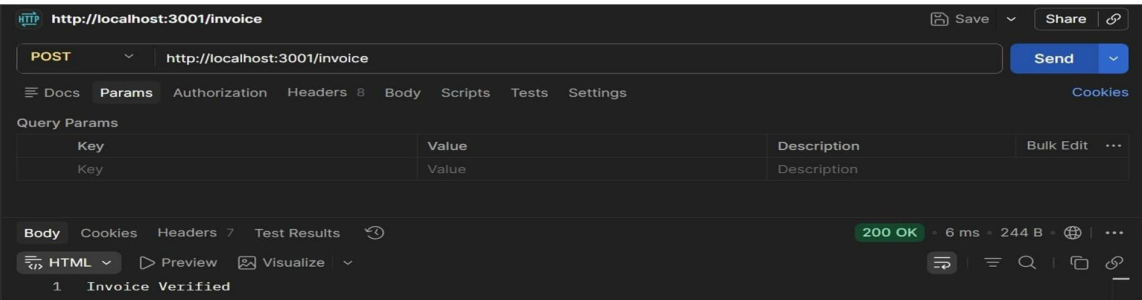
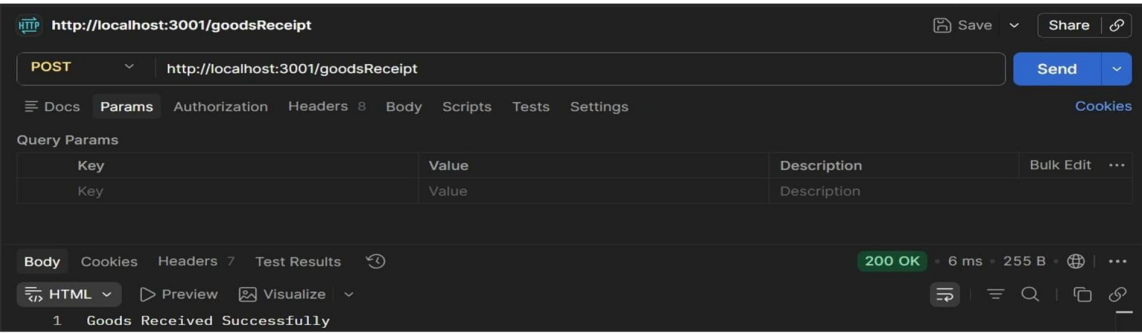
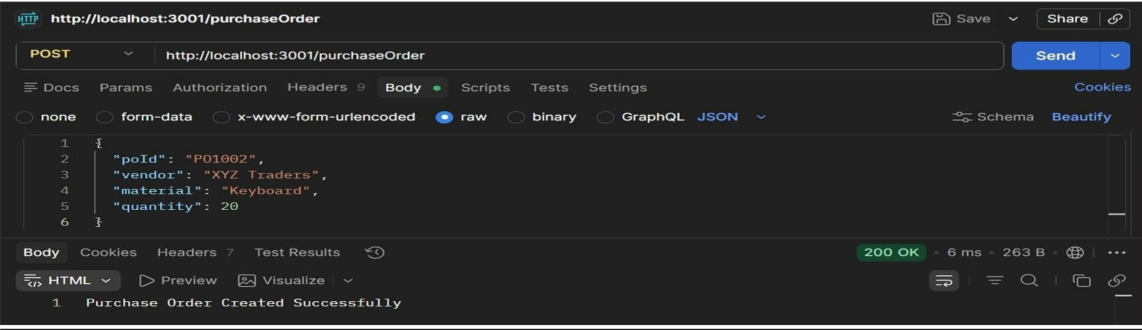
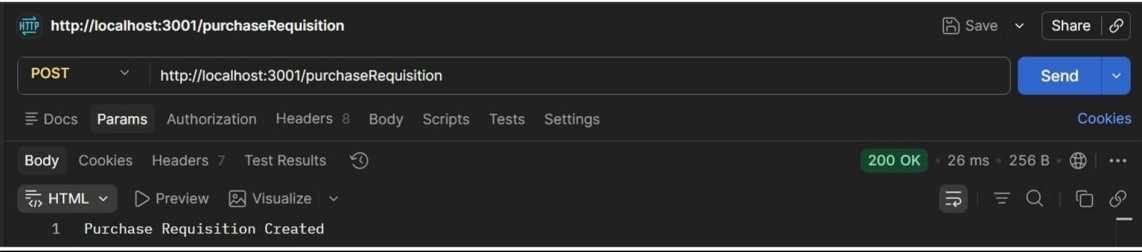
```
JS server.js x {} vendor.json {} package.json {} purchaseOrder.json
JS server.js > ...
1  const express = require('express');
2  const app = express();
3  const vendors = require('./vendor.json');
4  const purchaseOrders = require('./purchaseOrder.json');
5
6  app.use(express.json());
7
8  // Step 1 - Get Vendors
9  app.get('/vendors', (req, res) => {
10    res.json(vendors);
11  });
12
13  // Step 2 - Create Purchase Requisition
14  app.post('/purchaseRequisition', (req, res) => {
15    console.log("Purchase Requisition Created");
16    res.send("Purchase Requisition Created");
17  });
18
19  // Step 3 - Create Purchase Order
20  app.post('/purchaseOrder', (req, res) => {
21    const newOrder = req.body;
22    purchaseOrders.push(newOrder);
23    console.log("Purchase Order Created");
24    res.send("Purchase Order Created Successfully");
25  });
26
```



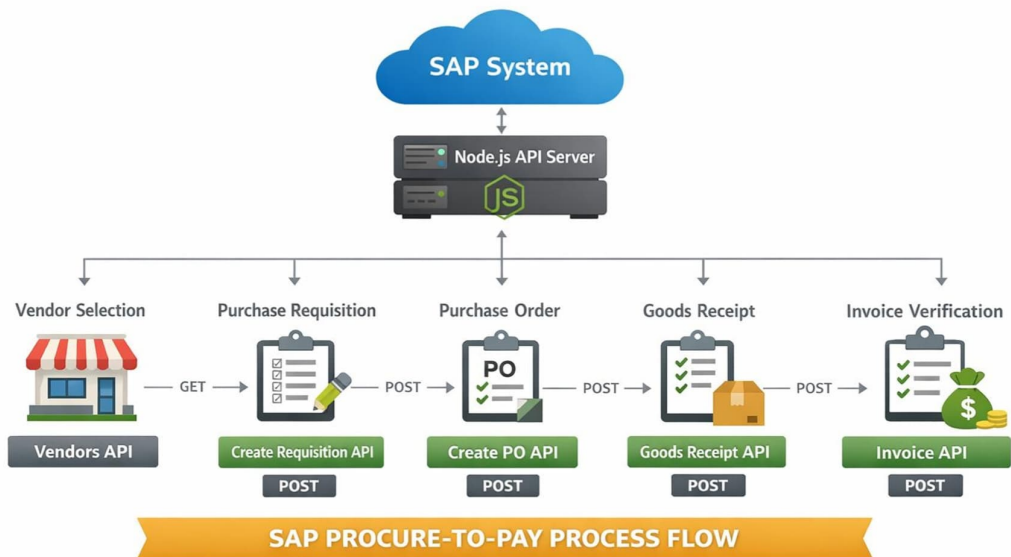
```
27  // Step 4 - Goods Receipt
28  app.post('/goodsReceipt', (req, res) => {
29    console.log("Invoice Verified");
30    res.send("Goods Received Successfully");
31  });
32
33  // Step 5 - Invoice Verification
34  app.post('/invoice', (req, res) => {
35    console.log("Invoice Verified");
36    res.send("Invoice Verified");
37  });
38
39  // Step 6 - Payment to Vendor
40  app.post('/payment', (req, res) => {
41    console.log("Payment Completed");
42    res.send("Payment Completed");
43  });
44
45  app.listen(3001, () => {
46    console.log("SAP P2P Integration Running on port 3001");
47  });
```



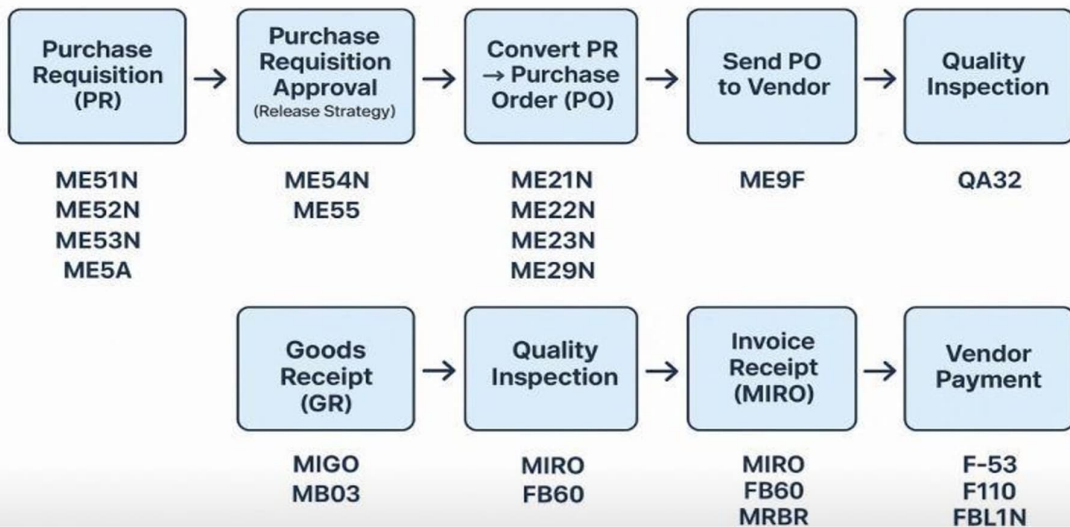
```
← → ↺ http://localhost:3001/vendors
JSON Raw Data Headers
Save Copy Collapse All Expand All Filter JSON
0:
  vendorId: "V001"
  vendorName: "ABC Supplier"
  city: "Hydrabad"
1:
  vendorId: "V002"
  vendorName: "XYZ Traders"
  city: "Delhi"
```



■ P2P workflow:



SAP P2P (Procure-to-Pay) Process



TECH STACK USED:

- Node.js
- Express.js
- JSON
- VS Code
- Postman

UNIQUE POINTS:

- Simulates SAP Integration without SAP access
- End-to-End P2P Process
- REST API based architecture
- Easy to understand and scalable
- Real-world business process simulation

FUTURE IMPROVEMENTS:

The project can be improved further by:

- Adding database integration
- Connecting to SAP Cloud Platform
- Creating UI dashboard
- Adding authentication and authorization
- Implementing error handling

CONCLUSION :

The SAP Procure-to-Pay Integration Project successfully demonstrates the procurement lifestyle using REST APIs. This project provides a practical understanding of SAP integration concepts and business workflow automation.

The system simulates real-world SAP processes and helps in learning SAP Integration Developer role.